

APV

PI 435 ~~Section~~ (1) CP 10317

Harvesting

* 1.1 Harvesting of the following crops.

(1) Cassava — When matured, Cassava is harvested by carefully uprooting of the tuber using manual or mechanical method. Whilst ever the system used, it is very important that the uprooting be done with care, otherwise the operations can cause breaking or mechanical damage on the tuber or Cassava, thus reducing the products quality and fastening subsequent losses from the action of mould and bacteria.

1-2 *

* Processing of Cassav tubers.

After harvesting, Cassava roots are very susceptible to spoilage, and without any preservation measures, can only be stored for about 48 hours before they begin to deteriorate. Therefore the root must be processed

(1)

As soon as possible after harvest to arrest—the physiological process and the subsequent deterioration.

After harvest, Cassava has to be peeled washed, using 7 clean water, then grate dehydrate the grated Cassava and sieved and let it ~~ferment~~ using 9/10 oil or Palm oil, and ~~inspiration~~ is ~~inspired~~ forced to reduce the moisture content. And if the gas is not well dried, it can be re-dried before Packaging into bags.

13 #ansing of Rice

One of the simplest system of ^{harvesting} rice is by cutting the rice stem 1m above the soil level with the sheaf attached to the stem, and mostly 11- is divisible to 4 in the morning and 1st cutting of the rice stem and sheaf is still wet & ^{will not} be used.

(3)

And this has to be pulled up on a
trampling or clean ones to avoid
contamination. And rice is to pick up
with the sheaf space and strike the
panicles against a hard surface. Another
frequent method of threshing rice is to
trample it under foot. It can also be done
by striking sheaves repeatedly on a threshing
floor with a stick. The threshing floors
on which the sheaves are spread must
have a hard clean surface. By using
of the above method of hand treading,
a worker can obtain 15 to 40 kg of
product per hour.

WINNOWER

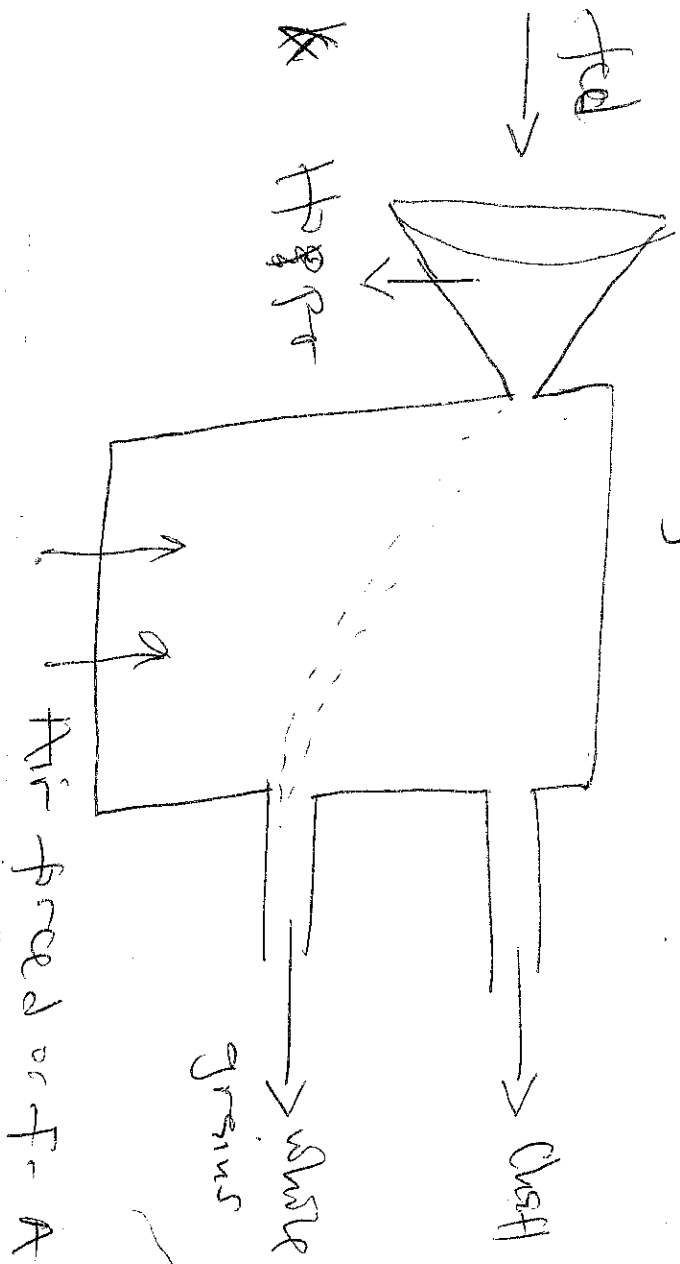
8.75

Winnowing is a method of separating
the grains from the chaff, it can be done
manually by lifting the grains about
your shoulder forwards the direction
of air. The speed of the air will carry
the chaff away and leaving the grains

to fall on the trompeling.

(4)

And also it can be done mechanically by using a winnowing machines, which is based on the differences in the terminal velocity of grain and chaff. Raw material (grain) is fed into a cylindrical column as shown in the diagram below. Air is forced up the column against the grains & chaff. The speed of the air is such that it allows the grains to fall through a discharge gate while the lighter chaff is blown away.



(3)

The whole grain will be properly dried, graded and packaged in form of flour.

1.4. The basic operation of farm processing equipment.

- Q1
1. Combine harvester
 2. Corn sheller *
 3. Rice thresher and
 4. Milling machine. *

(1) Combine harvester:— Combine harvester has an important place in the post-harvest production process, especially for the convertibility.

By the simple replacement of a few accessories and appropriate changes in settings, these machine can treat different kinds of grains (e.g. rice, maize, beans, wheat, sorghum, sunflowers, soybean, groundnut etc) equipped with a rotating threshing drum. (with beater or teeth) these machine often have a device to shake out straw and to clean the grain.

(6)

a. Corn Sheller - Manual sheller, which are relatively common and some times made by local artisans, permit easier and faster shelling of ears of maize. These come in different models, some of them equipped to take a motor; they are generally driven by a handle or a pedal. Use of manual shellers generally requires only one worker with yields from 14 to 100 kg, they are well adapted to the needs of small scale production. Mechanical sheller, shellers self propelled or tractor-driven, these sheller are often mounted on rubber-tired wheels for easy movement to the field. The use of motorized sheller may require two or more workers. Yields depend on the type of machine, the nature and maturity of the grains, the skill of the worker and organization of the work, and they can vary from 100 to 5000 kg.

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~~B~~

(7)

Rice thresher :- Threshing of rice can be done using different methods:-
Hand threshing :- by striking the panicles against a hard surface or a sample is under foot and by beating the sheaves on a threshing floor with a stick.

Threshing with vehicles :- This operation ~~is~~ is also called treading out, can especially well be accomplished with vehicles. Tractor can also be used, by running the tractor twice over, sheaves of rice that are spread in layers on a circular threshing floor. The sheaves must be turned over by the two passages of the tractor and the fields of about 640 kg can be obtained.

Threshing with hand-driven machines :-
Machines driven by a manual device are often used to improve fields

(8)

and working condition during threshing.

By means of the handle, a big drum fitted with metal nicks or teeth is made to rotate. The rice is threshed by hand holding the sheaves and pressing the panicles against the rotating drum. The speed of the threshing drum must be kept at about 300 revolutions per minute (rpm).

The hand held sheaves must all be of the same length with the panicles all laid in the same direction, and the grains must be very ripe and dry. The machine may be confusingly and regularly fed, but without introducing excessive moisture of product.

If the paddy obtained contains too many unthreshed panicles and plant residues, a second threshing must be followed by an effective cleaning of the product.

4. Milling machines — Milling is the breaking up of product into smaller particles as a result of compact. There are different devices used in milling, these are
1. Traditional method of milling.
 2. Modern method of milling.

(1) Under the traditional method of milling the method of mortar and pestle is the most widely used in the West African country like, Benin, Nigeria, Ghana, Senegal, Sierra Leone, Liberia, Ivory Coast, etc. In these countries, women sit on the ground or on a low stool and pound the grains in a mortar. This method allows the homogenization of the mixture inside the grains.

During the milling, the pounding mortar is interrupted by several sittings that allow to obtain a range of different products. The advantage of this method is its versatility. The woman can control the operation so that she can adapt according to requirement available.

Another method of milling and the traditional method is by rubbing the two stones to grind.

(2) Modern method of milling

Very Hammer mill machine: The hammer mill machine consists essentially of a rotor carrying a roller that bears a number of hammer round its periphery as shown in the diagram below.

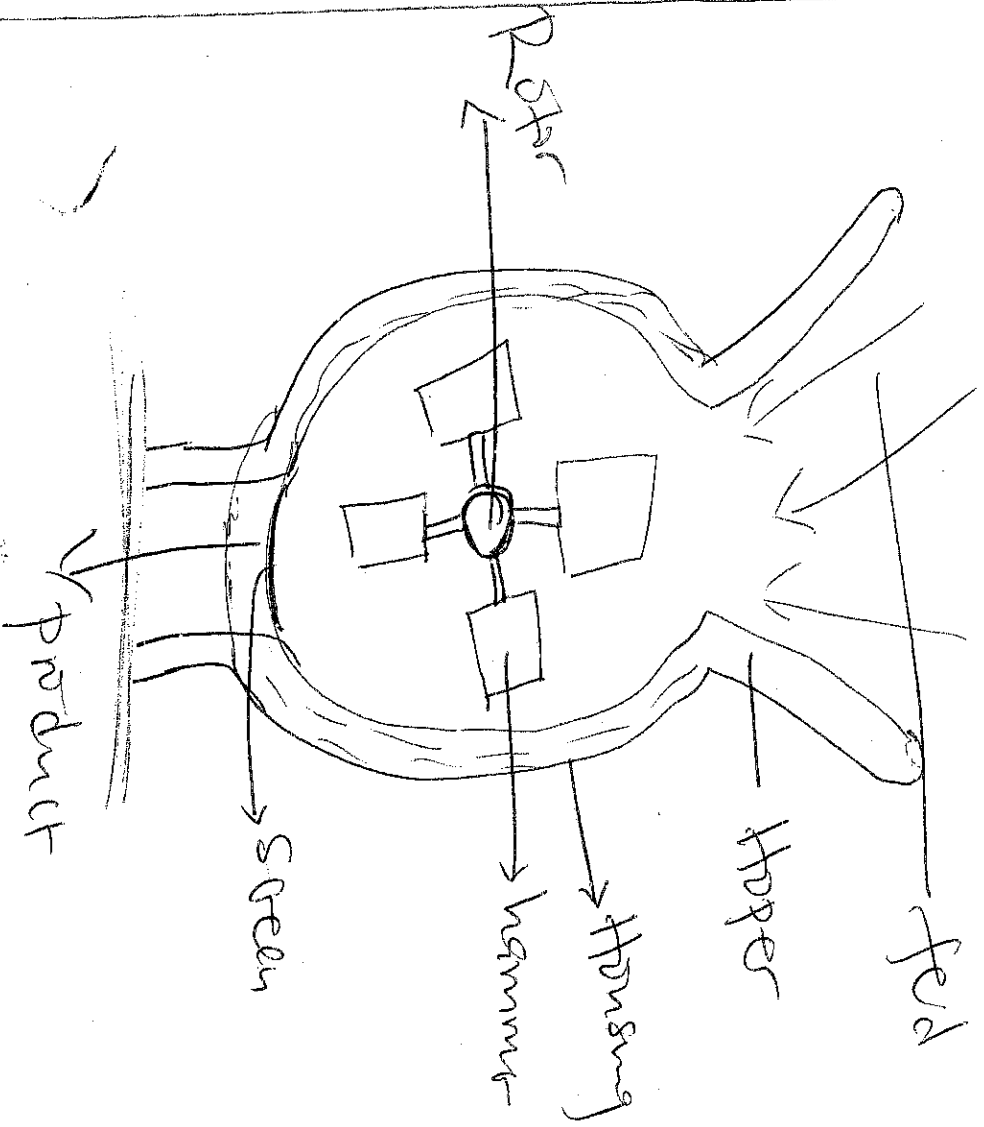
The housing is circular but it can also be rectangular. A break plate is used to enhance comminution. Water & steam is used to recycle the material between the required size of the product.

The size of the product can be changed with by varying rotor speed, feed rate, clearance b/w hammers and grinding plate, number in size of hammers and the size of the screen. The mill is good for hard crystalline solid, fibrous materials and is used extensively for paper pulp and manufacturing.

(ii) ^{dis} Advantages of the mill includes high power requirement and non uniform grinding.

However the disadvantage includes Versatility, freedom from damage by foreign materials.

Hammer mill machine



Drying & Storage facilities (12)

Drying

Drying is another good aspect of agricultural activities. ~~However~~ ^{before} ~~seed moisture~~ ^{seed moisture} limit should be done rather quickly. If the seeds are to be artificially dried, they should be supplied to processing plants soon after harvesting; if preliminary drying is to be done at the farm, the seed should be evenly spread in thin layers either on floors or on tarpaulins.

Methods of Drying

1) sun drying and (2) forced air drying.

(1) Sun drying - In the absence of forced drying the moisture content of seeds have to be reduced in field before harvest and later by sun drying on the threshing floor. The system involves harvesting & crops when they are fully dried in the field, leaving the harvested produce in the field for a couple of days to sun-dry and later spreading the threshed and unmowed produce in thin layers on harvesting floors to sun-dry.

(Continue in part 15)

Storage facilities (13)

1- Maize Cobs:- Cobs are, for the most part, wooden structures used for the storage of both dehusked and un dehusked maize, unthreshed Sorghum, Paddy and unshelled Grain legumes.

The floor of the storage compartment is 1m above the ground level and is supported by vertical poles, each measuring 1.5m. There are a varying number of long vertical poles providing support for the products by rats, metal guards are fitted to the vertical support below the cob floor level.

The width of the cob is 60cm for humid areas but could be as much as 150cm for more arid environments. It may be as high as required in each situation so that Cobs look like giant baskets. The sides are ventlated and are usually made of bamboo, grass strick or even wire netting and the entire cob is located in such a way that the prevailing winds blow perpendicular to the length. The construction of the Cobs is completed with a provision of a thatched or metal roof.

2) Bins — Plastic and metal drums capable of holding 150 kg of grains with perfect-fitting lids which have been thoroughly cleaned, and which ~~have~~ do not have perforations are now being used as air-tight containers for the preservation of grain. By fitting them to the top oxygen is excluded from the mass and this will prevent orthopedic Pest from developing.

3) Earthen Silos or Chambers granaries:

Earthen Silos or Chambers as they are known in Northern Nigeria, are constructed with ordinary clay, which is sometimes mixed with fragments of plant material or with bricks made with clay. They are protected from dampness by raising them from the ~~protected~~ from ground by means of wooden poles, clay or large stones.

They vary widely in shape and size. They are sometimes covered with a straw mat or built completely of mud without a roof, the opening being on the side or on top. They are a cheap means of storage well suited to dry climates. They also provide an effective means of protection against Pest since they can be sealed hermetically.



(15)

If sun-drying is done to be done, the following precautions should be taken to ensure seed quality.

1. Do not spread the produce on wet, dirty, or wetting floor.
2. Only one crop variety, and produce from one plot, should be bunched on a threshing floor, in order to maintain lot-identity and to avoid mechanical admixture.

(2)

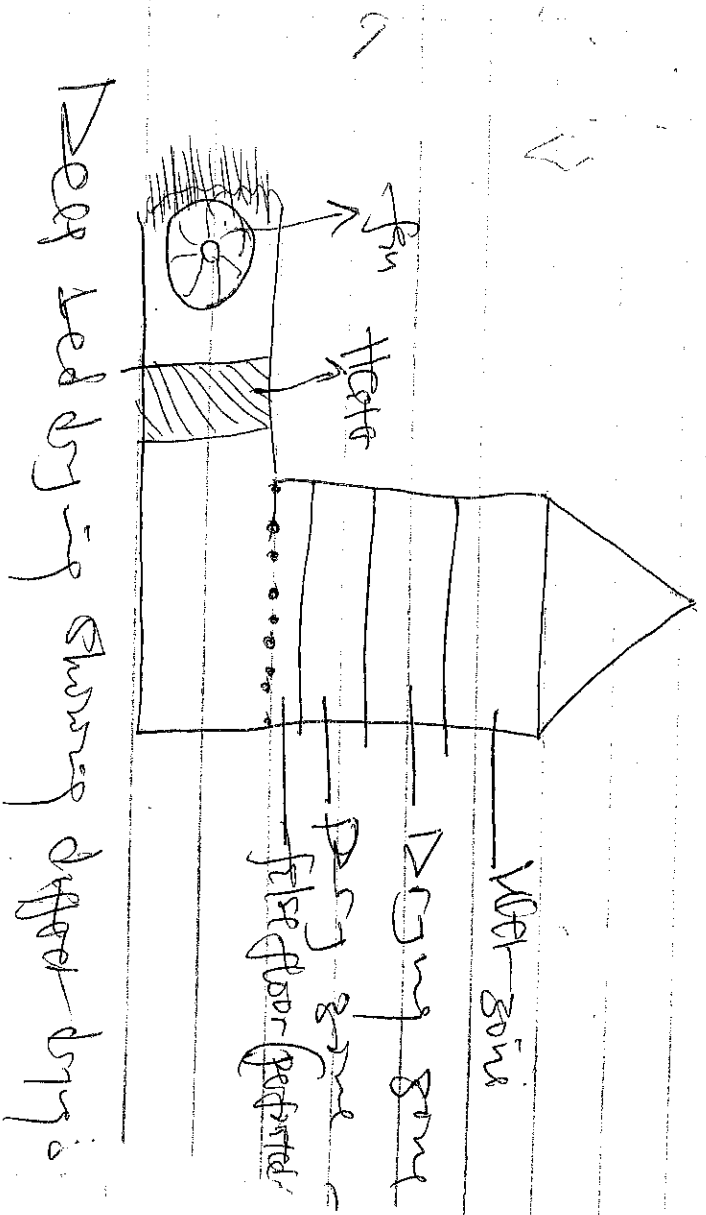
Forced air drying:- In this system, air (natural or heated) is forced into seeds. The air passing through damp seeds picks up water. The evaporating coils the air & the seed.

There are different types of driers, as listed below.

1. Redrying - Drying by sprinkling from a redrying e.g. microwave oven.
2. Freeze drying - moisture in this product is frozen by refrigeration.
3. Deep bed drier - In this type of drier, in which the material stacked in relatively deep bin such as a silos as shown below.

(16)

Usually cylindrical structure, at the top at located fan or blower, a heater, a Plenum (open space) and supported floor. Heated air is forced through the product and is allowed to escape from the top. Result of the relatively high depth involved drying is more uniform. Rather than any point in time there is a dry zone, drying zone and a wet zone. And this is controlled by moisture migration as air moves through the drying system.



(17)

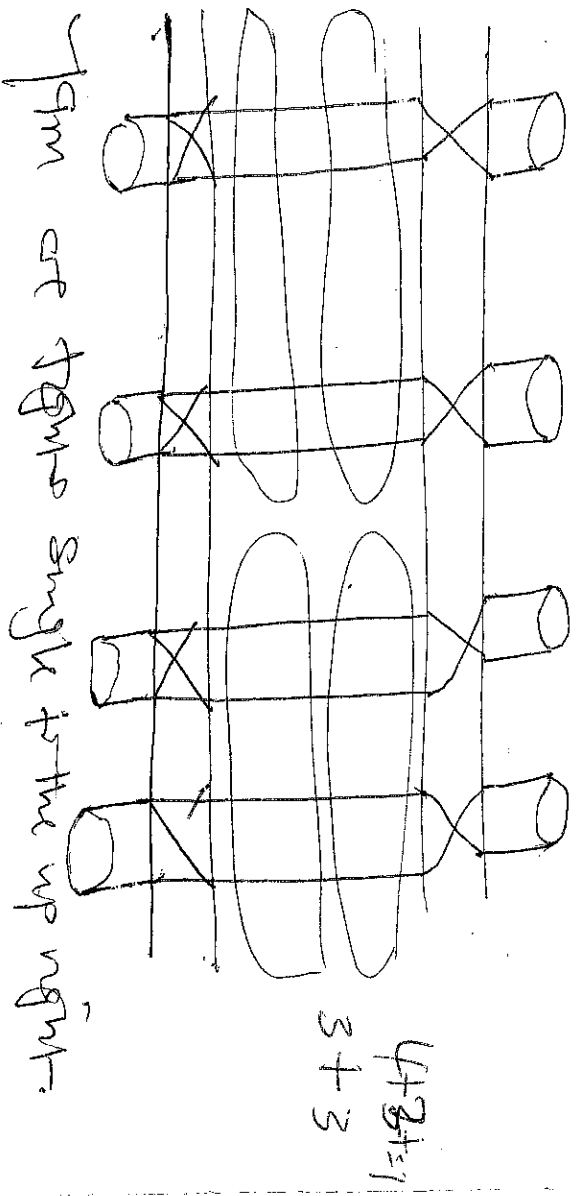
Construction of barn, nest, Gals, and Plumbers :-

1. - Jam barns :- In hot, wet forest areas, jam are stored in barns. The jams may be carried by birds put in the barns. The important features of these barns are :-

- (i) Good ventilation of air round the jam.
- (ii) Shade from the heat of the sun.
- (iii) Security against pest.

A strong wall is made by planting posts, such as fast-growing trees. The post sprout branches and leaves which shade.

Jam barn



Plumbus (18)

Storage of cereals and pulses. Once a Cereal crop is harvested it may have to be stored for period of time before it can be marketed or used as feed or seed. The length of time Cereals can be safely stored will depend on the conditions in which it is harvested and the type of storage facility being utilized. Proper conditions to store grains effectively are those which prevent or discourage the growth of micro-organisms and insect. Such conditions involves Control & maintenance

of

- Temp of grains

- Temp moisture content of grains
- condition and soundness of grain

- Oxygen supply to the storing environment.

✓ * There are different types of storage facilities of cereals & grains, these are

- (i) bins or metal bin
- (ii) silos
- (iii) Pouches.
- (iv) bags etc